Multimedia Signal Processing - 2nd module

Prof. Marco Tagliasacchi

September, 27 2011

1 Question - 10 pts

Consider a source X that emits symbols from a binary alphabet a, b. Let $p(i) = Pr\{X = i\}$, $p(i|j) = Pr\{X_k = i|X_{k-1} = j\}$. Encode the sequence of symbols $aaba$ by means of Elias coding, showing the corresponding bitstream. Consider the following three cases:

1. Assume that X is a discrete memoryless source, with $p(a) = 0.8$
2. Assume that X is a discrete source with memory, with $p(a) = 0.8$, and
   - $p(a|a) = 0.9$
   - $p(a|b) = 0.3$
3. Assume that X is a discrete source with memory, with $p(a) = 0.8$. Conditional probabilities are to be estimated as encoding progresses.

Solution

1. Let $p(b) = 1 - p(a) = 0.2$. The unit interval is partitioned into two sub-intervals $[0, 0.8)$, $[0.8, 1)$. Encoding proceeds as follows:

symbol	interval	length
a	$[0, 0.8)$	0.8
a	$[0, 0.64)$	0.64
b	$[0.512, 0.64)$	0.128
a	$[0.512, 0.6144)$	0.1024

   Let us select a number in the interval $[0.512, 0.6144)$. For example, the midpoint is 0.5632. We need to retain the first $l = -\lceil \log_2 0.1024 \rceil + 1 = 5$ bits of the binary expansion.
   The binary expansion of 0.5632 can be found as follows:

0.5632	$\times 2$	1.1264	1	1/2
0.1264	$\times 2$	0.2528	0	0
0.2528	$\times 2$	0.5056	0	0
0.5056	$\times 2$	1.0102	1	1/16
0.0102	$\times 2$	0.0204	0	0

   The encoded bitstream is 10010.

2. Let $p(b) = 1 - p(a) = 0.2$, $p(a|a) = 0.9$, $p(b|a) = 0.1$, $p(a|b) = 0.3, p(b|b) = 0.7$. The unit interval is partitioned into two sub-intervals $[0, 0.8)$, $[0.8, 1)$. Then, the current interval is partitioned based on the observed symbols at the previous step

symbol	interval	length	p
a	$[0, 0.8)$	0.8	$p(a) = 0.8$
a	$[0, 0.72)$	0.72	$p(a|a) = 0.9$
b	$[0.648, 0.72)$	0.072	$p(b|a) = 0.1$
a	$[0.648, 0.6696)$	0.0216	$p(a|b) = 0.3$

Let us select a number in the interval $[0.648, 0.6696)$. For example, the midpoint is 0.6588. We need to retain the first $l = -\lceil \log_2 0.0216 \rceil + 1 = 7$ bits of the binary expansion.
The binary expansion of 0.5632 can be found as follows:

0.6588	$\times 2$	1.1376	1	1/2
0.1376	$\times 2$	0.2752	0	0
0.5504	$\times 2$	1.1008	1	1/8
0.1008	$\times 2$	0.2016	0	0
0.2016	$\times 2$	0.4032	0	0
0.4032	$\times 2$	0.8064	0	0
0.8064	$\times 2$	1.6128	1	1/128

The encoded bitstream is 1010001.

3. Let $p(b) = 1 - p(a) = 0.2$. We initialize two histograms, to keep track of the statistics. That is, $n(a|a) = k, n(a|b) = k, n(b|a) = k, n(b|b) = k$. For example, we set $k = 1$.

symbol	interval	length	$n(a\|a)$	$n(b\|a)$	$n(a\|b)$	$n(b\|b)$	p
a	$[0, 0.8)$	0.8	1	1	1	1	$p(a) = 0.8$
a	$[0, 0.4)$	0.4	2	1	1	1	$p(a\|a) = 1/(1+1) = 0.5$
b	$[0.266, 0.4)$	0.133	2	2	1	1	$p(b\|a) = 1/(2+1)2 = 0.333$
a	$[0.3226, 0.4)$	0.0666	2	2	2	1	$p(a\|b) = 1/(1+1) = 0.5$

Let us select a number in the interval $[0.3226, 0.4)$. For example, the midpoint is 0.3613. We need to retain the first $l = -\lceil \log_2 0.0666 \rceil + 1 = 5$ bits of the binary expansion.
The binary expansion of 0.5632 can be found as follows:

0.3613	$\times 2$	0.7226	0	0
0.7226	$\times 2$	1.4552	1	1/4
0.4552	$\times 2$	0.9104	0	0
0.9104	$\times 2$	1.8208	1	1/16
0.8208	$\times 2$	1.6416	1	1/32

The encoded bitstream is 01011.

2 Question - 5 pts

Illustrate the block diagram of a lossy image coding architecture based on transform coding (both encoder and decoder).

Solution

See slide set `MMSP_3.1_Image coding`.

3 Questions - 5 pts (each answer can be either TRUE or FALSE)

1. Let X denote a continuous source, and $D(R)$ the corresponding rate-distortion function, and σ_x^2 its variance.
   - [T] [F] $H(X) = \frac{1}{2} \log \frac{\sigma_x^2}{D}$
     FALSE. The entropy is undefined for continuous valued sources.
   - [T] [F] $R_1 > R_2 \Rightarrow D(R_1) \geq D(R_2)$
     FALSE. The rate-distortion function $D(R)$ is monotonically non-increasing.
   - [T] [F] $D(R) = \epsilon \sigma_x^2 2^{-2R}$
     TRUE. The rate-distortion function can be expressed as above, where ϵ depends on the p.d.f.
2. Consider a speech coding system
   - [T] [F] Coding intelligible speech typically requires at least 32 kbps
     FALSE. Intelligible speech might require less than 10 kbps.
   - [T] [F] The perceptual filter $W(z)$ weights more the error in the formant regions of the spectrum
     FALSE. Error in formant regions is less perceivable. Hence, it is weighted less by the perceptual filter.
   - [T] [F] Voiced speech can be synthesized based on the spectral envelope only
     FALSE. In addition to the spectral envelope, the fine periodic structure of the spectrum related to pitch needs to be recovered.
   - [T] [F] The computational complexity of the encoder in DPCM is smaller than in analysis-by-synthesis vocoders
     TRUE. In a DPCM scheme, the encoder only needs to compute prediction residuals. In the case of A-DPCM, complexity might be similar, as in both cases the encoders need to estimate the spectral envelope (i.e. the coefficients of the whitening filter $A(z)$).
3. Consider a video coding system and a GOP structure $IPP...I$
   - [T] [F] The coding efficiency improves by increasing the GOP size
     TRUE. P-frame require fewer bits than I-frames, at the same target quality.
   - [T] [F] The computational complexity increases by increasing the GOP size
     TRUE. P-frame are more complex to encode, due to the impact of motion estimation.
   - [T] [F] The adoption of P-frames introduces drift
     FALSE. Drift is avoided by encoding prediction residuals according to a closed-loop DPCM scheme.

4 Question - 12 pts

Load the signal x (`load speech`), which contains a speech fragment quantized at 8 bits per sample.

1. Consider the signal as a memoryless source that emits symbols in [-128:127]. Which is the minimum rate achievable if the signal is coded lossless?
2. Normalize x between -1 and 1 and compare the performance of PCM and DPCM lossy coding. Plot the rate-distortion curves for the two coding schemes. Let the number of bits per sample vary between 1 and 7 and express the rate in kbps.

Solution

```
clear all
close all
clc

%load signal
load speech;

%lossless coding
h = hist(x,-128:127);
p = h./sum(h);
p = p(p~=0);
H = -sum(p.*log2(p));
disp(['Entropy of the source is: ' num2str(H) ' bits/symbol']);
disp(['Minimum achievable rate: ' num2str(H*fs/1000) ' kbps']);

%normalize
x = x./max(abs(x));
%now signal range is between 1 and -1


%PCM
R = 1:7;
MSE_PCM = NaN*zeros(size(R));
ir = 1;
for r = R
    step = 2/2^r;
    x_pcm = step*floor(x./step) + step/2;
    index_pcm = floor(x./step);
    MSE_PCM(ir) = var(x-x_pcm);
    ir = ir+1;
end


%DPCM
%compute the error signal
e = x(1:end-1) - x(2:end);
range = max(e)-min(e);

x_dpcm = NaN*ones(size(x));
MSE_DPCM = NaN*zeros(size(R));
ir = 1;
for r = R
    step = range / 2^r;

    %first sample, PCM coding
    x_dpcm(1) = step*floor(x(1)/step) + step/2;

    for i = 2:length(x)

        e = x(i) - x_dpcm(i-1);
        eq = step*floor(e/step)+step/2;
        x_dpcm(i) = x_dpcm(i-1) + eq;

    end

    MSE_DPCM(ir) = var(x-x_dpcm);
    ir = ir+1;
end

%plot results
rate = (fs*R)/1000;
plot(rate,MSE_PCM,'--ob','Markersize',6)
hold on
plot(rate,MSE_DPCM,'--xr','Markersize',6)
ylabel('MSE');
xlabel('Rate [kbps]');
legend('PCM','DPCM');
```